
Data Analysis and Insights: Technology Survey

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OUTLINE



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EXECUTIVE SUMMARY



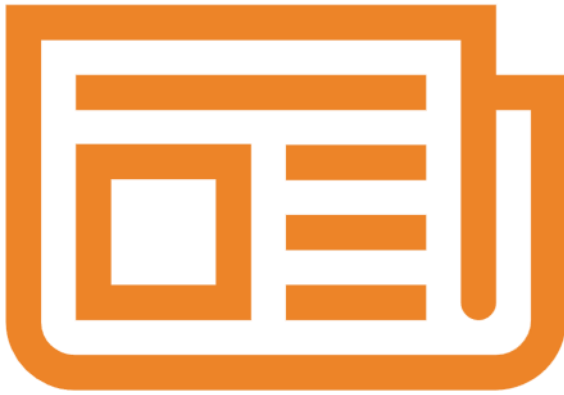
- JavaScript, SQL, and Python dominate today's technology stack.
 - The most widely used languages are JavaScript, SQL, HTML/CSS, TypeScript, and Python, confirming that modern software is built around interactive applications backed by data-driven systemsPoint2
- Developers are planning for more scalable, cloud-based systems.
 - PostgreSQL leads all databases today and continues to grow next year, while Redis and SQLite gain popularity, showing demand for both scalable databases and high-speed data access
- Framework choices reflect a move toward full-stack and web-centric development.
 - Node.js, React, and Next.js lead both current and future frameworks, indicating strong demand for JavaScript-based full-stack development Point5
- The developer workforce is highly educated and early-to-mid career.
 - Over 68% of respondents are between ages 25–44, and most hold at least a Bachelor's or Master's degree, showing a skilled and professionally mature developer population

INTRODUCTION



- This project analyzes global developer trends using data from the Stack Overflow Developer Survey and job posting datasets.
- The goal is to identify which programming languages, databases, platforms, and frameworks are most used today and which are expected to grow in the future.
- Multiple data sources were combined, including developer survey responses and real-world job market data.
- The analysis focuses on four major areas:
 - Programming languages
 - Databases
 - Cloud platforms
 - Web frameworks
- The insights from this project help reveal how technology trends are shaping the future of software development and data-driven applications.

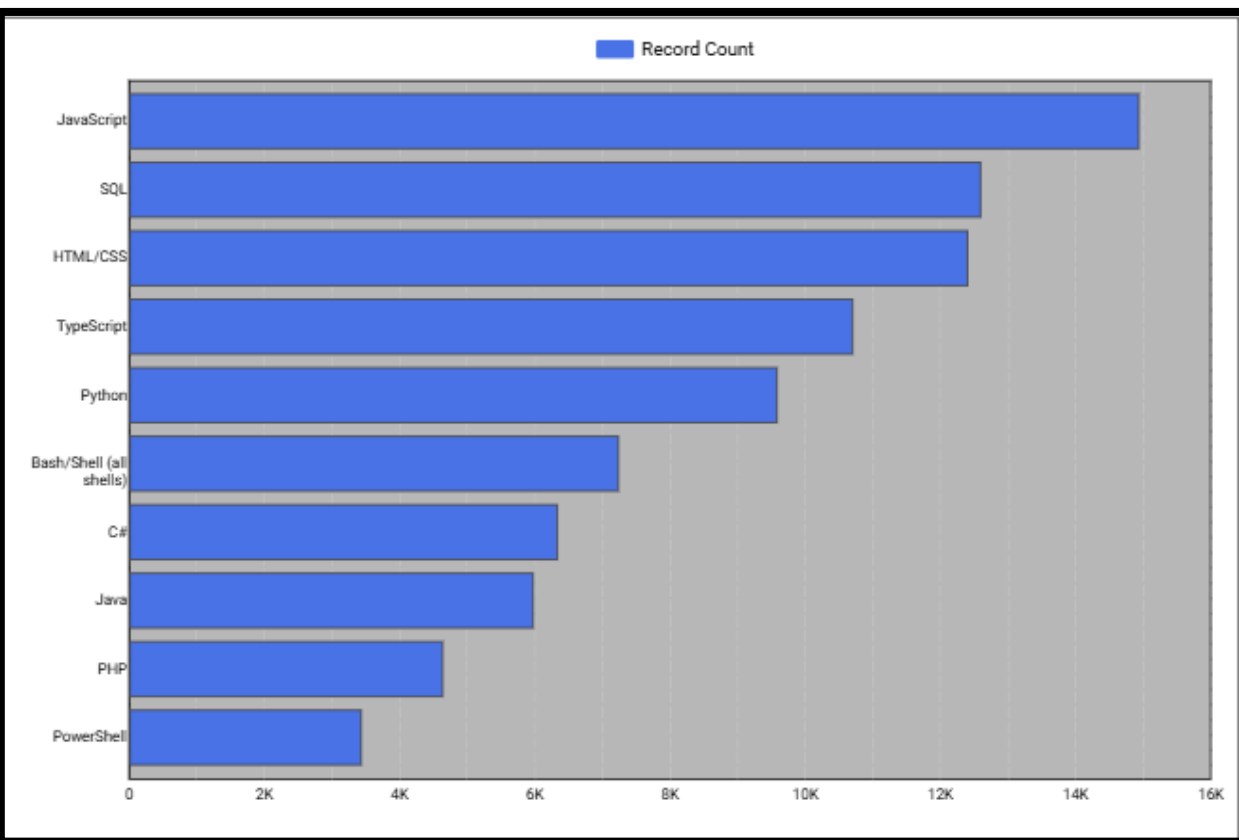
METHODOLOGY



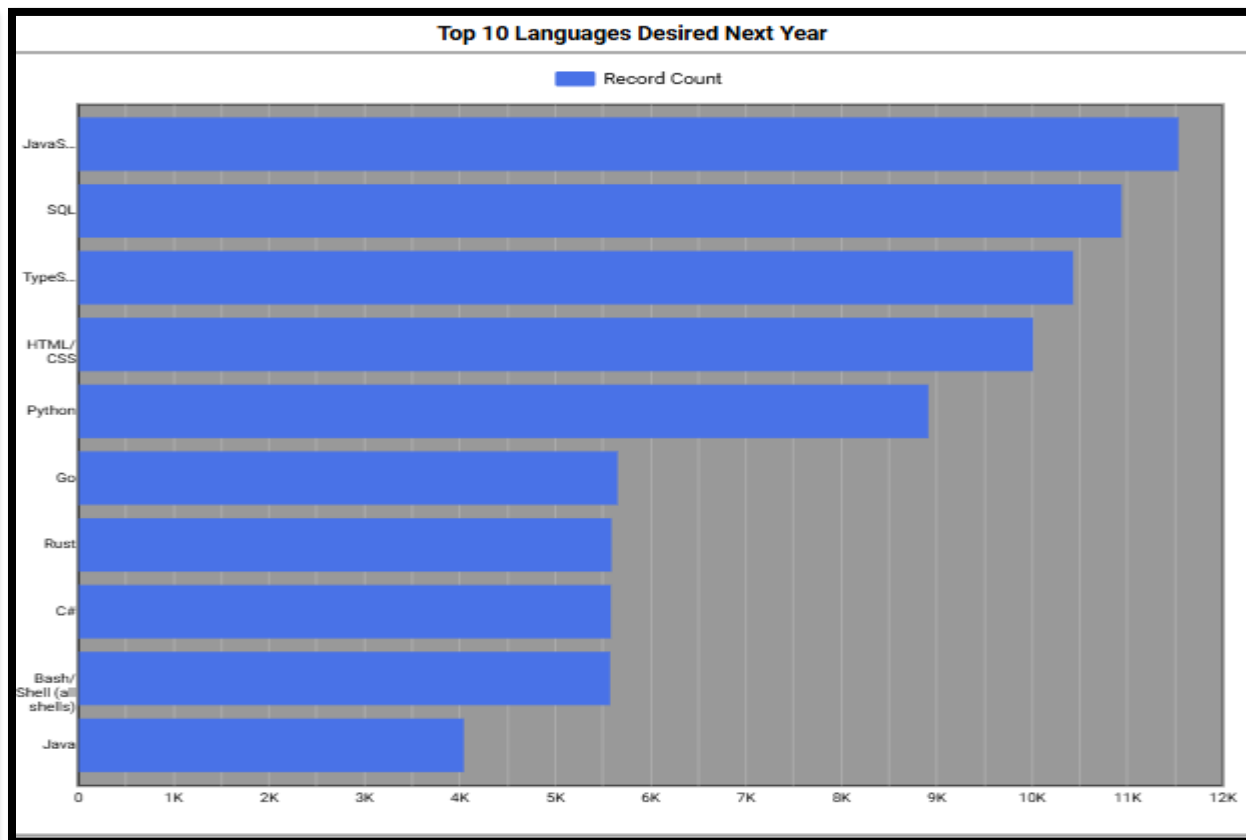
- Data was collected from multiple sources, including the Stack Overflow Developer Survey and job posting datasets obtained through APIs and web scraping.
- Raw datasets were cleaned and prepared using Python, including handling missing values, standardizing categories, and filtering relevant records.
- Data was analyzed to identify the most used and most desired technologies across programming languages, databases, platforms, and web frameworks.
- Aggregations and transformations were performed to compute record counts and rankings.
 - Top 10 technologies were selected based on usage and future interest.
 - Data was grouped and sorted to compare current trends with projected trends.
- Visualizations were created to communicate insights, including bar charts, bubble charts, word clouds, and geographic maps.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

1. JavaScript and SQL dominate both the current and projected next year.
2. Python remains strong in both years but shows slower growth compared to emerging languages
3. TypeScript is gaining momentum and shows clear future growth.

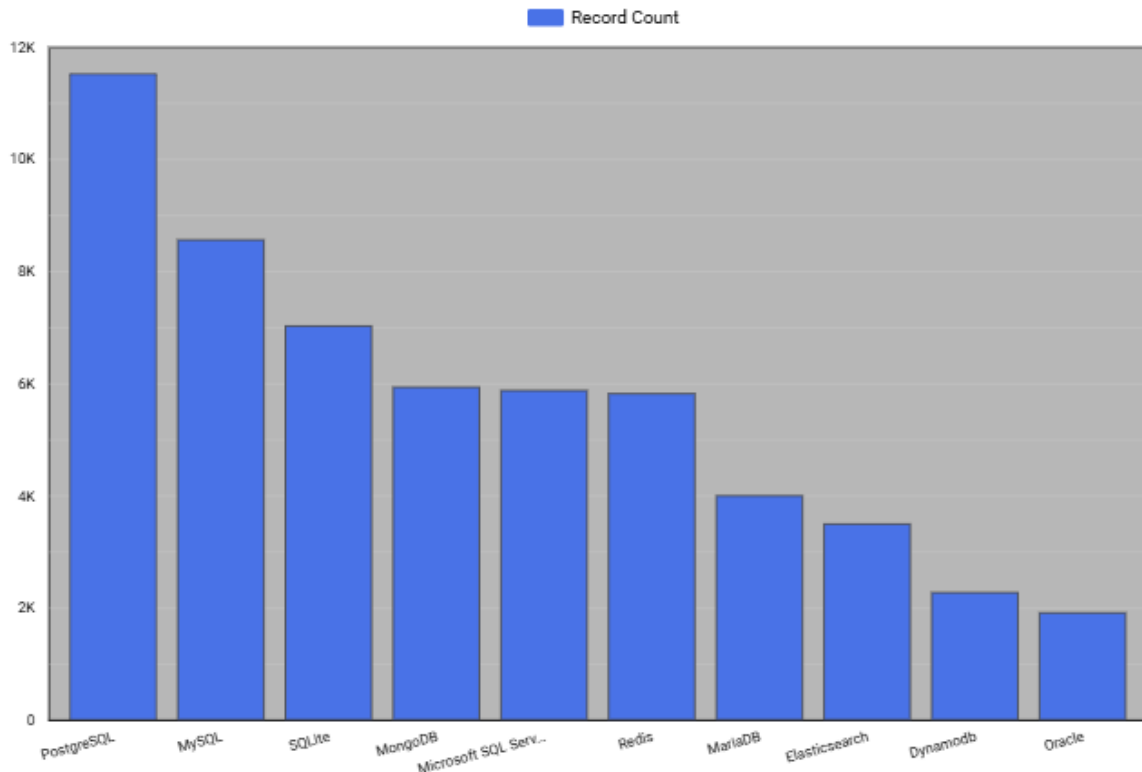
Implications

1. JavaScript and SQL form the backbone of modern data-driven applications, connecting user interaction directly to live data
2. Python is now a foundational skill rather than a competitive differentiator, as newer technologies take the spotlight.
3. The rise of TypeScript reflects a shift toward larger, more reliable, and more secure software systems

DATABASE TRENDS

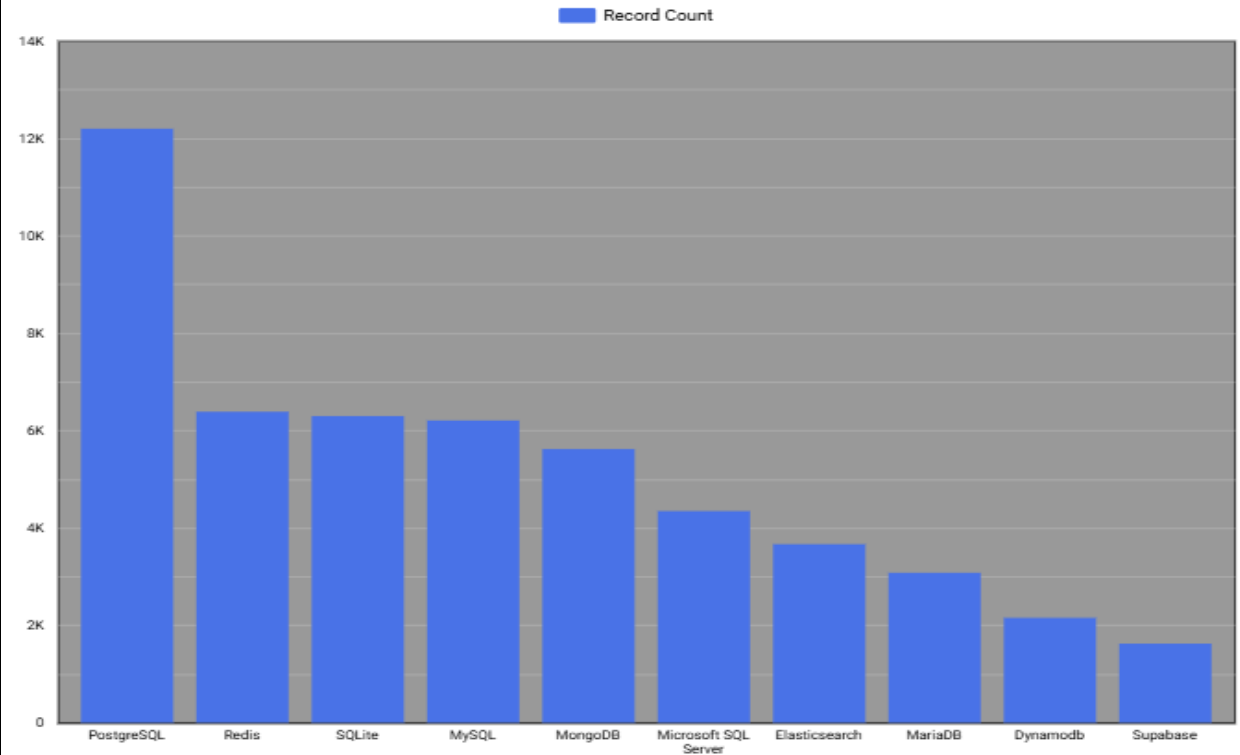
Current Year

Top 10 Databases Used



Next Year

Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most widely used database and continues to grow into next year
- Traditional SQL databases continue to dominate the database landscape
- SQLite and Redis show increased interest in the “Next Year” chart, indicating rising demand for fast, lightweight, and flexible data storage

Implications

- PostgreSQL’s growth suggests companies want a single, versatile database that supports analytics, transactions, and scaling
- SQL remains the foundation of modern data systems, even as new technologies emerge
- The rise of Redis reflects the need for high-speed data access to support real-time and AI-driven applications

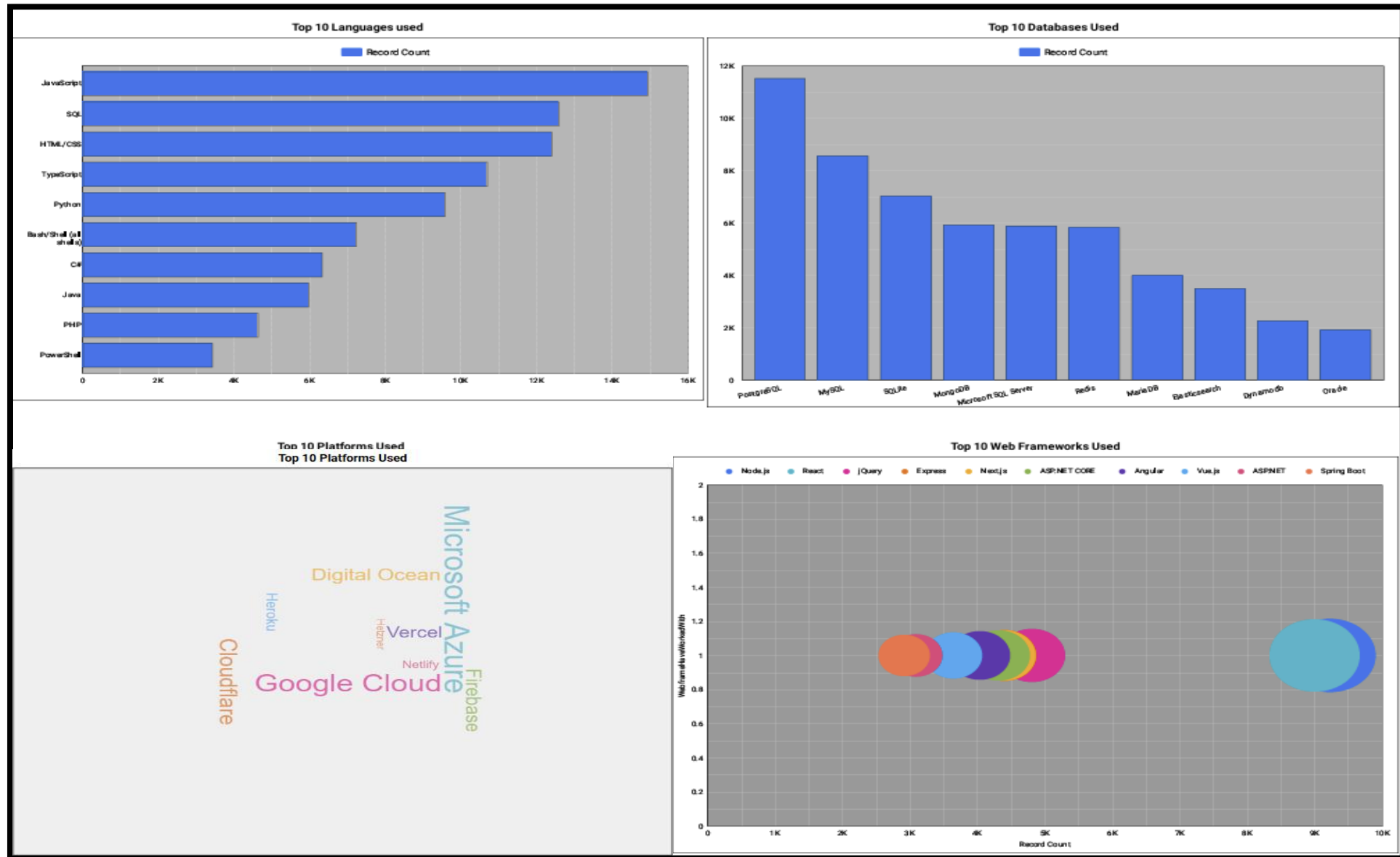
DASHBOARD



To view dashboard go to:

<https://lookerstudio.google.com/reporting/5d4d53a7-a4e0-4d14-b1ce-db76882f6a57>

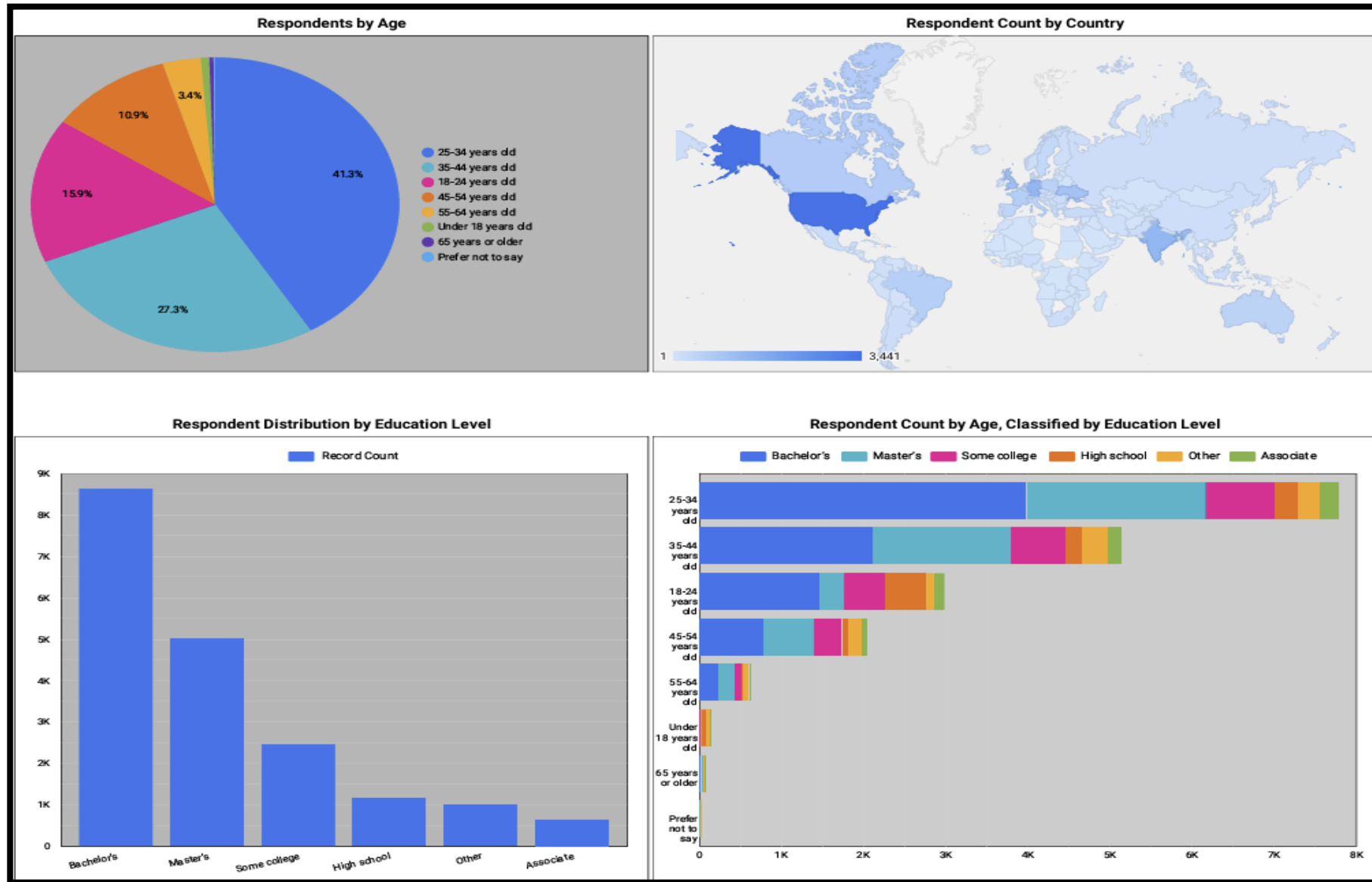
DASHBOARD TAB 1: Current Technology Usage



DASHBOARD TAB 2: Future Technology Trends



DASHBOARD TAB 3: Demographics



DISCUSSION



- The results show a strong alignment between what developers use today and what they want to use in the future, indicating stability in the core technology stack.
- JavaScript, SQL, and PostgreSQL remain dominant because they support the growing demand for interactive, data-driven, and scalable applications.
- The increasing interest in TypeScript, Redis, and modern web frameworks reflects a shift toward building larger, more complex, and more performance-focused systems.
- Cloud platforms continue to grow because modern software increasingly depends on distributed, cloud-based infrastructure.
- These trends suggest that future software development will be shaped by the need for speed, scalability, real-time data processing, and AI-driven applications.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Full-stack, data-driven development dominates the industry. JavaScript, SQL, Python, and modern web frameworks lead both usage and future demand.
- The technology stack is shifting toward scalable cloud-based systems. PostgreSQL, Redis, and cloud platforms are the foundation of modern applications.
- Developers are prioritizing performance, reliability, and real-time data. The rise of TypeScript, Redis, and modern frameworks reflects a move toward larger, more complex, and more dependable systems.

Implications

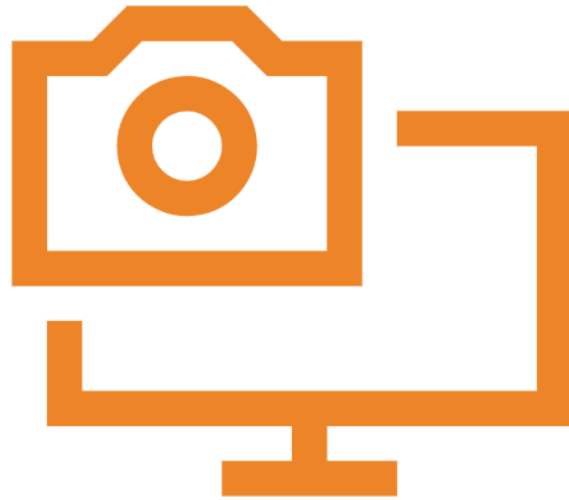
- Successful developers must combine software engineering with data skills. Modern roles increasingly require both front-end interaction and backend data management.
- Cloud and database technologies are becoming as important as programming languages. Understanding platforms like AWS and databases like PostgreSQL is now critical for building and scaling applications.
- Future growth will be driven by AI, analytics, and real-time systems. Technologies optimized for speed, scalability, and intelligent data processing will continue to gain importance.

CONCLUSION



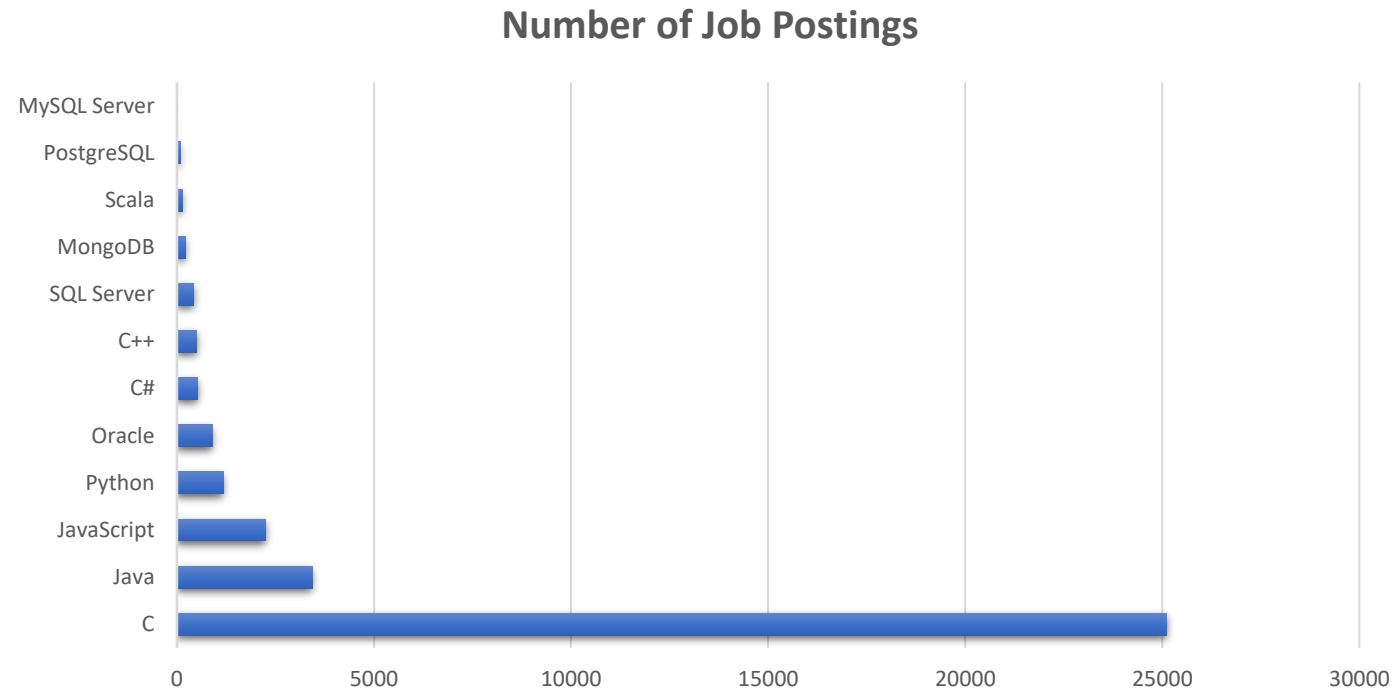
- The analysis shows that modern software development is driven by data-centric, cloud-based, and full-stack technologies.
- JavaScript, SQL, Python, and PostgreSQL remain the core tools powering today's and tomorrow's applications.
- The growing adoption of cloud platforms, modern frameworks, and high-performance databases reflects the industry's shift toward scalable and real-time systems.
- These trends indicate that the future of technology will be shaped by AI, analytics, and cloud-native development, making data-driven skills increasingly valuable.

APPENDIX



- Relevant additional charts, or tables are in the Appendix.

JOB POSTINGS



POPULAR LANGUAGES

